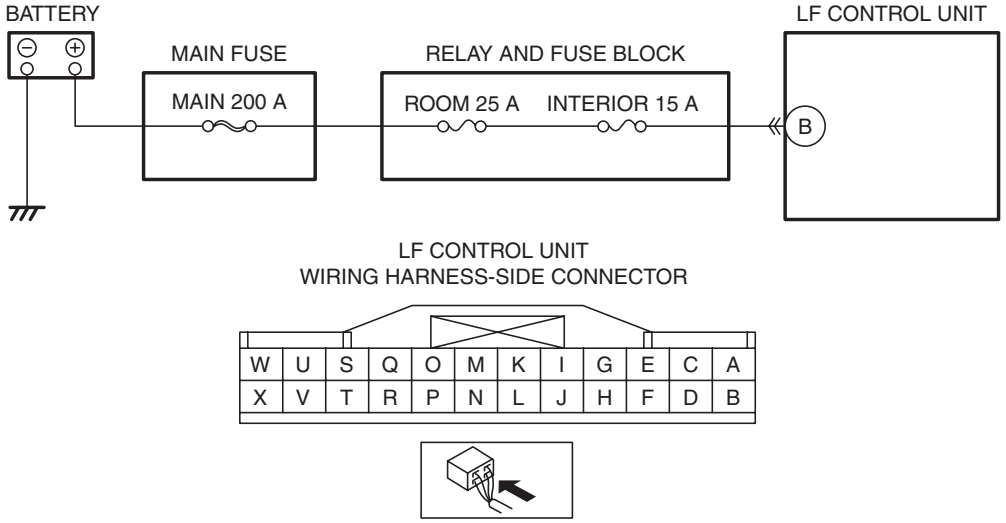
DTC B13C3:16 [ADVANCED KEYLESS ENTRY SYSTEM]

id0902p8027100

Without i-ELOOP

Description	LF control unit power supply voltage low input
Detection condition	• Start stop unit power supply circuit voltage of **8.5 V or more or less than 16.5 V** and LF control unit power supply circuit voltage of **5 V or more or less than 8.5 V** are detected for **5 s or more**.
Fail-safe	• Inhibits the door lock/unlock control using the advanced keyless entry system.
Possible cause	• LF control unit connector or terminal malfunction • Malfunction between battery positive terminal and LF control unit terminal B • LF control unit malfunction • Start stop unit malfunction



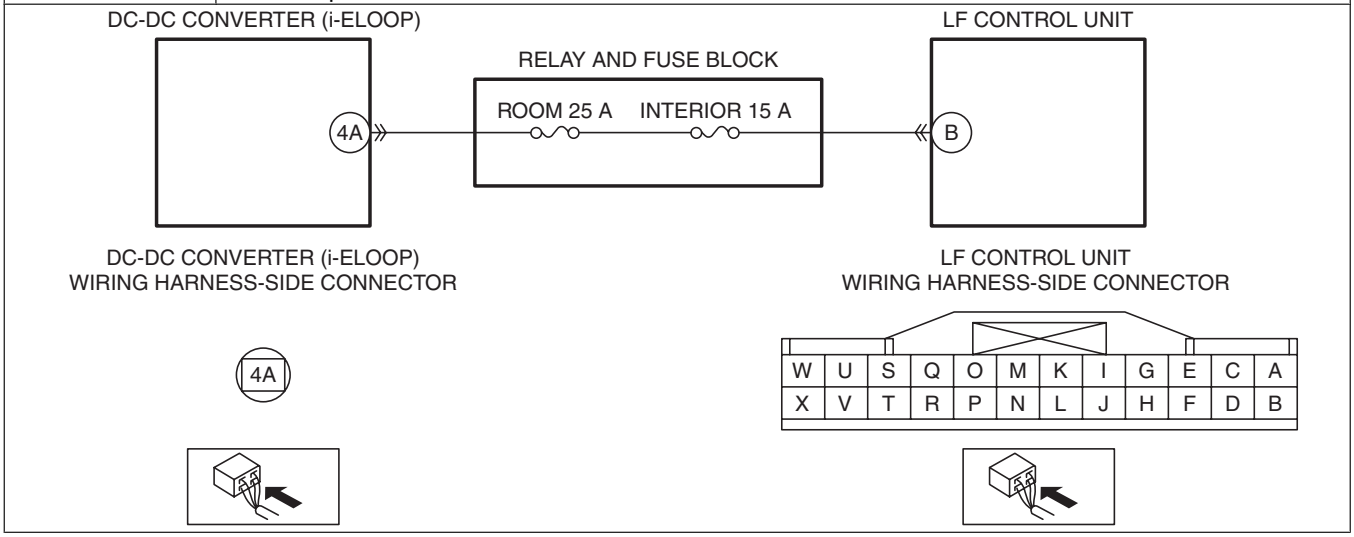
Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT LF CONTROL UNIT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the LF control unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 4.
2	**VERIFY LF CONTROL UNIT POWER SUPPLY VOLTAGE** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the LF control unit terminal B (wiring harness-side). • Is the voltage **5 V or more or less than 8.5 V**?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and LF control unit terminal B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a malfunction. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect the wiring harness between battery positive terminal and LF control unit terminal B. — If there is any malfunction: • Repair or replace the wiring harness. — If there is no malfunction: • Replace the LF control unit. (See LF CONTROL UNIT REMOVAL/INSTALLATION.) Go to Step 4.
		No	Go to the next step.

Step	Inspection		Action
3	**INSPECT LF CONTROL UNIT** • Inspect the LF control unit. (See LF CONTROL UNIT INSPECTION.) • Is the LF control unit normal?	Yes	Go to the next step.
		No	Replace the LF control unit, then go to the next step. (See LF CONTROL UNIT REMOVAL/INSTALLATION.)
4	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the advanced keyless entry system using the M-MDS. (See CLEARING DTC [ADVANCED KEYLESS ENTRY SYSTEM].) • Switch the ignition ON (engine off) and wait for **5 s or more**. • Retrieve the advanced keyless entry system DTCs using the M-MDS. (See DTC INSPECTION [ADVANCED KEYLESS ENTRY SYSTEM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
5	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ADVANCED KEYLESS ENTRY SYSTEM].)
		No	DTC troubleshooting completed.

With i-ELOOP

Description	LF control unit power supply voltage low input
Detection condition	• Start stop unit power supply circuit voltage of **8.5 V or more or less than 16.5 V** and LF control unit power supply circuit voltage of **5 V or more or less than 8.5 V** are detected for **5 s or more**.
Fail-safe	• Inhibits the door lock/unlock control using the advanced keyless entry system.
Possible cause	• DTCs are stored in the DC-DC converter (i-ELOOP). — DC-DC converter (i-ELOOP) connector or terminal malfunction — DC-DC converter (i-ELOOP) malfunction • LF control unit connector or terminal malfunction • Malfunction between DC-DC converter (i-ELOOP) terminal 4A and LF control unit terminal B • LF control unit malfunction • Start stop unit malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY DC-DC CONVERTER (i-ELOOP) DTCs** • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.
2	**INSPECT LF CONTROL UNIT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the LF control unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 5.
3	**VERIFY LF CONTROL UNIT POWER SUPPLY VOLTAGE** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the LF control unit terminal B (wiring harness-side). • Is the voltage **5 V or more or less than 8.5 V**?	Yes	Disconnect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and LF control unit terminal B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a malfunction. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect the wiring harness between battery positive terminal and LF control unit terminal B. — If there is any malfunction: • Repair or replace the wiring harness. — If there is no malfunction: • Replace the LF control unit. (See LF CONTROL UNIT REMOVAL/INSTALLATION.) Connect the service plug, then go to Step 5. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
		No	Go to the next step.
4	**INSPECT LF CONTROL UNIT** • Inspect the LF control unit. (See LF CONTROL UNIT INSPECTION.) • Is the LF control unit normal?	Yes	Go to the next step.
		No	Replace the LF control unit, then go to the next step. (See LF CONTROL UNIT REMOVAL/INSTALLATION.)
5	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the advanced keyless entry system using the M-MDS. (See CLEARING DTC [ADVANCED KEYLESS ENTRY SYSTEM].) • Switch the ignition ON (engine off) and wait for **5 s or more**. • Retrieve the advanced keyless entry system DTCs using the M-MDS. (See DTC INSPECTION [ADVANCED KEYLESS ENTRY SYSTEM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.

Step	Inspection		Action
6	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ADVANCED KEYLESS ENTRY SYSTEM].)
		No	DTC troubleshooting completed.